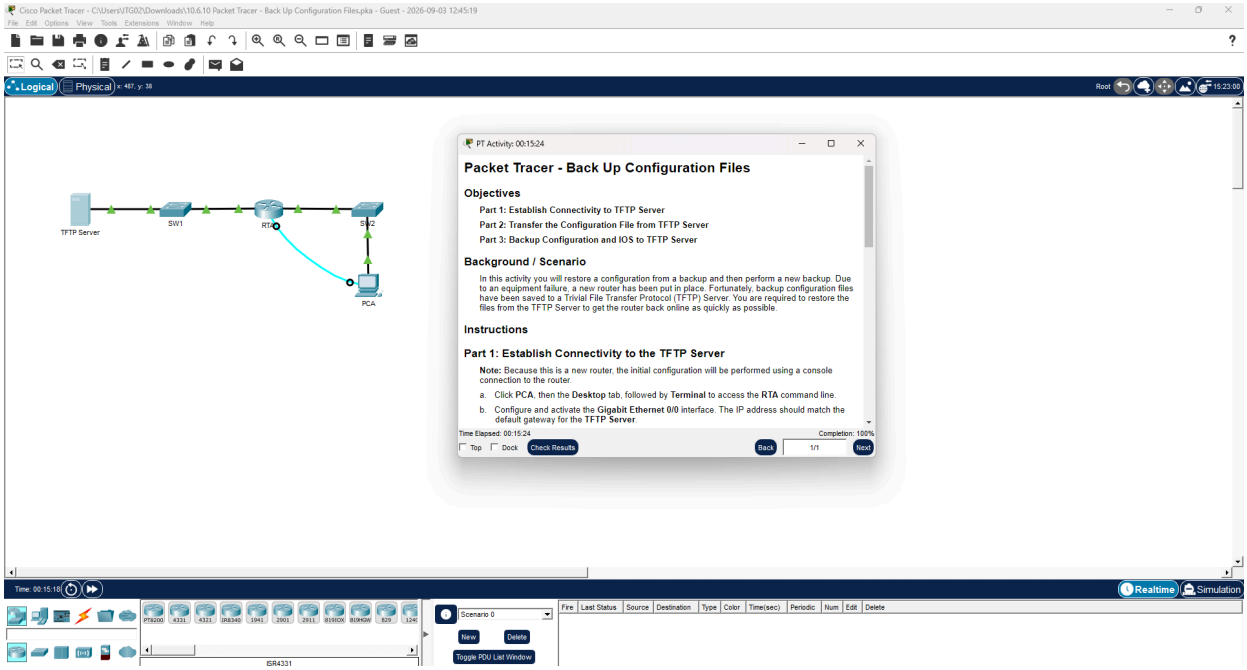


1. Executive Summary & Objectives

Due to a physical hardware failure on the primary edge router, a replacement Cisco router was installed in the topology. This activity demonstrates how to restore network operations by downloading saved configuration files from a Trivial File Transfer Protocol (TFTP) server, performing critical interface adjustments, updating system identification, and executing full backup routines for both system configurations and Cisco IOS images.

Key Lab Objectives

- **Part 1:** Establish initial console connectivity and enable network links to the TFTP Server.
- **Part 2:** Restore configuration files from the TFTP Server to active memory (RAM) and resolve interface status issues.
- **Part 3:** Perform routine backup operations by exporting running configurations and the Cisco IOS binary image to the TFTP Server.

2. Network Topology & Addressing Table

Topology Description

- **PCA:** Connected via Console cable to **RTA** (Rollover cable to Console Port) and via Ethernet cable to interface `GigabitEthernet 0/1`.

- **RTA / RTA-1:** Central 1900-series Cisco Router routing traffic between LAN host (PCA) and Server Segment.
- **TFTP Server:** Dedicated storage server connected directly to `GigabitEthernet 0/0` on **RTA**.

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
TFTP Server	NIC	172.16.1.2	255.255.255.0	172.16.1.1
RTA / RTA-1	GigabitEthernet0/0	172.16.1.1	255.255.255.0	N/A
RTA / RTA-1	GigabitEthernet0/1	172.16.1.254	255.255.255.0	N/A
PCA	NIC	172.16.1.225	255.255.255.0	172.16.1.254
PCA	Console	Serial (RS-232)	N/A	N/A

3. Step-by-Step Implementation & Device Commands

Part 1: Establish Connectivity to the TFTP Server

1. Accessed router **RTA** via **PCA Terminal** over console connection.
2. Configured IP address on `GigabitEthernet0/0` matching the default gateway required for TFTP communication.

```
None
Router> enable
Router# configure terminal
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 172.16.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
```

3. **Connectivity Verification:** Tested IP reachability from RTA to the TFTP server (`172.16.1.2`).

```
None
Router# ping 172.16.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
```

Part 2: Transfer Configuration File from TFTP Server

1. Downloaded saved configuration `RTA-confg` directly into active `running-config` memory.

```
None
Router# copy tftp: running-config
Address or name of remote host []? 172.16.1.2
Source filename []? RTA-confg
Destination filename [running-config]?

Accessing tftp://172.16.1.2/RTA-confg...
Loading RTA-confg from 172.16.1.2: !
[OK - 785 bytes]

785 bytes copied in 0.001 secs
RTA#
%SYS-5-CONFIG_I: Configured from console by console
```

2. Checked interface states following configuration merge:

```
None
RTA# show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     172.16.1.1      YES manual up                    up
GigabitEthernet0/1     172.16.1.254    YES TFTP   administratively down   down
```

3. Interface Correction & Activation: Re-enabled `GigabitEthernet0/1` interface to establish PCA LAN connection.

t

RTA# configure terminal

RTA(config)# interface GigabitEthernet0/1
RTA(config-if)# no shutdown
RTA(config-if)# end
RTA#

None

Part 3: Back Up Configuration and IOS Image to TFTP Server

1. Renamed host device to `RTA-1` and committed RAM settings to NVRAM.

```text
RTA(config)# hostname RTA-1
RTA-1# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

2. Exported updated running-config to TFTP Server as RTA-1-confg.

t
RTA-1# copy running-config tftp:
Address or name of remote host []? 172.16.1.2
Destination filename [RTA-1-confg]?

!!
[OK - 810 bytes]

810 bytes copied in 0.012 secs

None

3. Inspected flash filesystem to verify Cisco IOS image filename:

```text
RTA-1# show flash:
System flash directory:
File  Length   Name/status
  3  33591768 c1900-universalk9-mz.SPA.151-4.M4.bin
```

```
  2  2825     sigkey-20130611.pub
  1  1138     mime.819.pt
[33595731 bytes used, 221718829 available, 255314560 total]
1024K bytes of processor board System flash (Read/Write)
```

4. Transferred system IOS binary image to TFTP Server repository:

```
None
RTA-1# copy flash: tftp:
Source filename []? c1900-universalk9-mz.SPA.151-4.M4.bin
Address or name of remote host []? 172.16.1.2
Destination filename [c1900-universalk9-mz.SPA.151-4.M4.bin]?

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[OK - 33591768 bytes]

33591768 bytes copied in 2.451 secs
```

4. Lab Questions & Technical Analysis

Part 2, Question (b): What changes were made after issuing the `copy tftp running-config` command?

- **Answer:** Importing `RTA-confg` instantly merged saved settings into RAM:
  - System **hostname** changed automatically from `Router` to `RTA`.
  - User line access parameters (Console/VTY passwords) and encrypted privilege mode credentials were enabled.
  - IP addressing and interface descriptions for `GigabitEthernet0/1` (`172.16.1.254/24`) were populated.

Part 2, Question (c): Are all interfaces active?

- **Answer: No.** While `GigabitEthernet0/0` remained active, `GigabitEthernet0/1` was in an `administratively down` state. In Cisco IOS, network interfaces default to `shutdown` unless explicit `no shutdown` directives are applied. The restored

configuration script did not automatically bring the second physical interface online, requiring administrator intervention.

Part 3, Question (e): What special character repeatedly displays indicating that the IOS file is being copied to the TFTP server successfully?

- **Answer:** The **exclamation mark (`!`)** character. In Cisco IOS TFTP file transfers, each `!` indicates that a single UDP block/packet payload was acknowledged and successfully delivered to the remote host. (A period `.` would indicate a network timeout or dropped frame).

Part 3, Question (f): Has the IOS file `c1900-universalk9-mz.SPA.151-4.M4.bin` been copied to the TFTP Server?

- **Answer: Yes.** Navigating to **TFTP Server > Services > TFTP** confirms that `c1900-universalk9-mz.SPA.151-4.M4.bin` is listed among stored images. The file size matches the 33,591,768 bytes stored in router flash memory.

5. Summary Table & Key Technical Findings

Metric / Stage	Initial State	Final State	Technical Reason / Note
Device Hostname	`Router`	`RTA-1`	Initial restore restored `RTA`; updated manually to `RTA-1`.
G0/0 Interface	Down / Unconfigured	`172.16.1.1/24 (Up/Up)`	Configured manually to reach TFTP Server.
G0/1 Interface	Down / Unconfigured	`172.16.1.254/24 (Up/Up)`	Required `no shutdown` after restoring config.
Startup Configuration	Empty / Factory	Saved (810 Bytes)	Committed active settings via `copy run start`.
TFTP Backups	None (Replacement router)	`RTA-1-confg` & IOS Image	Verified on TFTP repository tab.

6. Recommendations for Production Environments

1. **NVRAM Verification:** Always execute `copy running-config startup-config` prior to backup transfers. Restoring a `running-config` without saving to startup memory leaves the router vulnerable to complete configuration loss upon reboot.
2. **Image Integrity Verification:** Cross-check file lengths using `show flash:` against file sizes on the TFTP Server after transfers to ensure binary files were not truncated during transmission.